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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110.

(An Autonomous Institution, Affiliated to Anna University, Chennai)

B.E. / B.Tech. End Semester Theory Examinations, November/December 2023.

Fifth Semester

Computer Science and Engineering

UCS2501 COMPUTER NETWORKS

(Regulations 2021)

Time: **Three Hours**

Maximum: **100 Marks**

K1: Remembering

K2: Understanding

K3: Applying

K4: Analyzing

K5: Evaluating

CO1:	To understand the protocol layering and physical level communication
CO2:	To understand the various components required to build different networks
CO3:	To learn the functions of network layer and the various routing protocols
CO4:	To familiarize the functions and protocols of the Transport layer
CO5:	To understand various application layer protocols

Part – A ($5 \times 2 = 10$ Marks)

		KL	CO	PI
1.	Define OSI model with a neat diagram.	K1	CO1	1.4.1, 13.3.1
2.	List the features and services of Data Link Layer.	K2	CO2	1.3.1
3.	What is meant by subnetting and supernetting?	K1	CO3	1.3.1
4.	State the purpose of port number and specify the port number for FTP.	K2	CO4	1.4.1, 13.3.1
5.	What is the purpose of Secure shell?	K1	CO5	1.3.1, 13.3.1

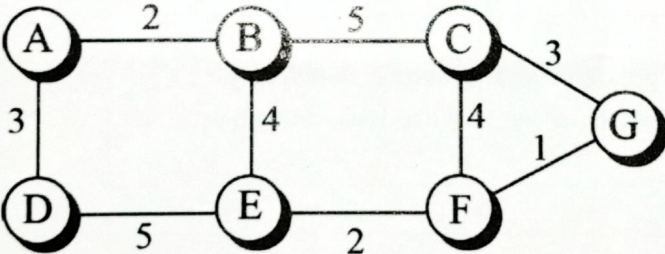
Part – B ($5 \times 6 = 30$ Marks)

		KL	CO	PI
6.	Explain about the Guided transmission Medias in computer networks?	K2	CO1	2.1.2, 13.3.1
7.	How the Carrier Sense Multiple Access Protocols are working?	K2	CO2	1.4.1, 2.3.1

8.	Explain about the classful addressing. Apply it to the following IP addresses to identify their classes. a) 200.11.8.3 b) 238.34.2.1 c) 114.34.2.8 d) 129.14.6.8 Illustrate the block diagram of the address 200.11.8.3 to which it belongs.	K3	CO3	1.4.1, 2.1.2
9.	Describe the three way handshake of TCP during connection establishment with flow diagram between client and the server with sequence number.	K2	CO4	1.4.1, 2.1.2
10.	Briefly discuss about the functionalities of SNMP and the PDU's exchanged between the SNMP manager and SNMP agent with proper illustration.	K3	CO5	2.3.1, 13.3.1

Part – C (5 × 12 = 60 Marks)

		KL	CO	PI
11.	When you write an email to your friend studying in Stanford University in USA, analyze how the email gets transferred through the underlying network. Make use of TCP/IP suite and map the layers of them based on their functionality.	K3	CO1	1.1.2, 1.3.1, 2.3.1, 2.4.1
(Or)				
12.	In your college, a new networking laboratory is established. As a network architect, analyze and state which topology could be applied for this scenario. Explain various types of network topologies with its advantages and disadvantages.	K3	CO1	1.1.2, 1.3.1, 2.3.1, 2.4.1
13.	Explain the technique which can be used for the following scenario. Identify the error correction technique and make use of it to encode the given data: 1010101. Find the redundant bits for encoding. During transmission, last LSB is corrupted. On the receiving end, detect the error and correct it.	K3	CO2	1.1.2, 2.1.2, 2.1.3, 2.4.1, 13.3.1
(Or)				

14.	a) Explain the media access protocol CDMA and apply it to the following scenario. Station A want to sends bit1, station B sends bit 0, station C sends bit 1 and station D remains silent. Make use of CDMA and find the data on the common channel. (8) b) List out the different types of Ethernet. In the standard Ethernet, with transmission rate of 10Mbps, we assume that the length of the medium is 2500m, propagation speed is 2×10^8 m/s and the size of the frame is 512 bits. Find the efficiency of Ethernet. (4)	K3	CO2	1.1.2, 2.1.2, 2.1.3, 2.4.1, 13.3.1
15.	Explain the distance vector algorithm. Apply this algorithm to the following network and build the initial routing table for the node C and then updated routing table after getting up update from B, F and G. (Note: Do not change the order) 	K3	CO3	1.1.2, 2.1.2, 2.1.3, 13.3.1
(Or)				
16.	An ISP is granted a block of addresses starting with 150.80.0.0/16. The ISP wants to distribute these blocks to 2600 customers as follows. a. The first group has 200 medium-size businesses; each need 128 addresses. b. The second group has 400 small businesses; each need 16 addresses. c. The third group has 2000 households; each need 4 addresses. Design the sub-blocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations.	K3	CO3	1.1.2, 2.1.2, 2.1.3, 13.3.1

17.	Draw the sender side FSM of TCP Tahoe. Apply TCP Tahoe algorithm to plot the congestion window graph versus RTT. The data transmission starts with $cwnd=1$, $ssthresh=16$ at $RTT=0$. At $RTT=1$, $cwnd=2$ and so on following TCP Reno principle. Consider the following sequence of events and plot the graph. - At $RTT=3$, the timeout occurs. - At $RTT=13$, three dupACK occurs. - Stop the graph at $RTT=20$	K3	CO4	2.1.2, 2.3.1, 2.4.1, 13.3.1
(Or)				
18.	Explain the different timers available in TCP. Apply Jacobson's RTO estimation and find the RTP by assuming, $RTT_m = RTT_s = 1.5$ $RTT_m = 2.5$, $RTT_s = 1.625$	K3	CO4	2.1.2, 2.3.1, 2.4.1, 13.3.1
19.	Identify the different protocols and technologies involved in accessing E-mail services. How can be those devised together?	K3	CO5	2.1.2, 2.3.1, 12.1.2 13.3.1
(Or)				
20.	Identify the protocol used to map domain name and IP address. Write and explain the two ways of resolving it with neat diagram and example.	K3	CO5	2.1.2, 2.3.1, 12.1.2 13.3.1